

Chapter 9b Practice Quiz

1. What are the first four nonzero terms in the Taylor series of $f(x) = e^{-4x}$ about $x = 0$?
2. Find the coefficient of x^3 in the Maclaurin polynomial for $f(x) = 3 \sin 2x$ at $x = 0$.
3. Use the second degree Maclaurin polynomial of $f(x) = \arctan x$ to approximate $f(0.2)$.
4. What is the third degree term of the Taylor polynomial, centered at $c = \frac{\pi}{6}$, for $f(x) = \cos(2x) + x$?
5. Write the Maclaurin series for f given $f(x) = \frac{\sin x}{x}$.
6. Which of the following is the Taylor series for $\frac{1}{x}$ about the point $c = 2$?

a) $\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{2^{n+1}}$ b) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} (x-2)^n}{2^{n+1}}$ c) $\sum_{n=0}^{\infty} \frac{(x-2)^n}{2^{n+1}}$ d) $\sum_{n=0}^{\infty} \frac{(x-2)^n}{2^n}$ e) $\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{2^n}$

7. Which of the following is equivalent to $1 + \frac{3}{1} + \frac{3^2}{2} + \frac{3^3}{6} + \frac{3^4}{24} + \frac{3^5}{120} + \dots$?

- a) 0 b) 2 c) $3e$ d) e^3 e) π

8. The function f has a Taylor series about $x=2$ and the n th derivative of f at $x=2$ is given by $f^{(n)}(2) = \frac{3^n}{(n+1)!}$ and $f(2)=1$. What is the second degree term of this series?

- a) $\frac{3}{2}x^2$ b) $\frac{3}{2}(x-2)^2$ c) $3(x-2)^2$ d) $\frac{3}{4}(x-2)^2$ e) $\frac{3}{4}x^2$

9. Which of the following is equivalent to $\sum_{n=0}^{\infty} \frac{(-1)^n x^{4n}}{3(2n)!}$?

- a) $\cos^2 x$ b) $\cos\left(\frac{x^2}{3}\right)$ c) $\frac{1}{3}\cos(x^2)$ d) $\sin\left(\frac{x^2}{3}\right)$ e) $\frac{1}{3}\sin(x^2)$

10. Let f be a function that has derivatives of all orders for all real numbers. Assume $f(0)=5$, $f'(0)=-3$, $f''(0)=1$, and $f'''(0)=4$.

a) Write the third-degree Maclaurin polynomial for f use it to approximate $f(0.2)$.

b) Write the third degree Taylor polynomial for h , where $h(x) = \int_0^{2x} f(t)dt$, about $x=0$.

c) A function $k(x)$ is defined as $k(x) = f(x) - \cos x + e^x$. Find the first 3 nonzero terms of the Maclaurin polynomial for $k(x)$.

Answers

1. Instead of creating this series, we can manipulate the known Maclaurin series for e^x to get to $f(x)$. So,

knowing $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$, we can replace the x with a $-4x$, giving us

$$e^{-4x} = 1 + (-4x) + \frac{(-4x)^2}{2!} + \frac{(-4x)^3}{3!} + \dots \text{ or } f(x) = e^{-4x} = 1 - 4x + 8x^2 - \frac{32}{3}x^3 + \dots$$

2. Instead of creating this series, we can manipulate the known Maclaurin series for $\sin x$ to get to $f(x)$.

So, knowing $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$, we can replace the x with a $2x$, giving us

$$\sin 2x = (2x) - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \dots \text{ So to get } f(x), \text{ we must multiply both sides by 3 or}$$

$$f(x) = 3 \sin 2x = 3(2x) - \frac{3(2x)^3}{3!} + \frac{3(2x)^5}{5!} - \dots \text{ Therefore the coefficient of the } x^3 \text{ term is found by}$$

$$\text{simplifying } -\frac{3(2x)^3}{3!} = -\frac{24x^3}{6} \text{ or } -4.$$

3. Since this is not one of the series we know, we must create the second degree Maclaurin polynomial from the beginning. So, a second degree Maclaurin polynomial is in the general form of

$$f(x) \approx P_2(x) = f(0) + \frac{f'(0)x}{1!} + \frac{f''(0)x^2}{2!}. \text{ So, } f(x) = \arctan x, f'(x) = \frac{1}{1+x^2}, f''(x) = \frac{-2x}{(1+x^2)^2} \text{ and}$$

$$f(0) = 0, f'(0) = 1, f''(0) = 0. \text{ Therefore, } P_2(x) = 0 + \frac{1x}{1!} + \frac{0x^2}{2!} \text{ or } f(x) \approx P_2(x) = x. \text{ So,}$$

$$f(0.2) = \arctan 0.2 \approx P_2(0.2) = 0.2.$$

4. Since this is not Maclaurin, we must create this polynomial, centered at $c = \frac{\pi}{6}$, from the general form of

$$f(x) \approx P_3(x) = f\left(\frac{\pi}{6}\right) + \frac{f'\left(\frac{\pi}{6}\right)}{1!}\left(x - \frac{\pi}{6}\right) + \frac{f''\left(\frac{\pi}{6}\right)}{2!}\left(x - \frac{\pi}{6}\right)^2 + \frac{f'''\left(\frac{\pi}{6}\right)}{3!}\left(x - \frac{\pi}{6}\right)^3. \text{ Since we only need to}$$

know the third degree term, we don't need to setup the rest. So,

$$f(x) = \cos 2x + x, f'(x) = -2\sin 2x + 1, f''(x) = -4\cos 2x, f'''(x) = 8\sin 2x \text{ and } f'''\left(\frac{\pi}{6}\right) = 4\sqrt{3}. \text{ So,}$$

$$\text{the third degree term is } \frac{4\sqrt{3}}{3!}\left(x - \frac{\pi}{6}\right)^3 = \frac{2\sqrt{3}}{3}\left(x - \frac{\pi}{6}\right)^3.$$

5. The Maclaurin series for f can be found by manipulating the Maclaurin series for $\sin x$, which is

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}. \text{ To get } f, \text{ we must divide each term by } x, \text{ which will}$$

$$\text{decrease each power by 1. This means } f(x) = \frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!}.$$

6. To determine the Taylor series, we should write out the first few terms and then try to establish a pattern. So the general form of the Taylor polynomial, centered about $c = 2$ is

$$f(x) = f(2) + \frac{f'(2)}{1!}(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3 + \dots \text{ So,}$$

$$f(x) = \frac{1}{x}, f'(x) = -\frac{1}{x^2}, f''(x) = \frac{2}{x^3}, f'''(x) = -\frac{6}{x^4} \text{ and}$$

$$f(2) = \frac{1}{2}, f'(2) = -\frac{1}{4}, f''(2) = \frac{1}{4}, f'''(2) = -\frac{3}{8}. \text{ Therefore, the first 4 terms of } f \text{ is}$$

$$f(x) = \frac{1}{2} + \frac{-1}{1!}(x-2) + \frac{1}{2!}(x-2)^2 + \frac{-3}{3!}(x-2)^3 + \dots \text{ or } f(x) = \frac{1}{2} - \frac{1}{4}(x-2) + \frac{1}{8}(x-2)^2 - \frac{1}{16}(x-2)^3 + \dots$$

This is alternating and the denominators are powers of 2, which works out to be **A**.

7. Looking at the denominators, we can see that these are all the factorials, so this is most likely related to

$$e^x. \text{ Since } e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \dots, \text{ we can see that the } x \text{ has been replaced with a 3.}$$

$$\text{Therefore, } 1 + \frac{3}{1} + \frac{3^2}{2} + \frac{3^3}{6} + \frac{3^4}{24} + \frac{3^5}{120} + \dots = e^3 \text{ or } \mathbf{D}.$$

8. The second degree term of a Taylor series centered at $x = 2$ is $\frac{f^{(2)}(2)}{2!}(x-2)^2$. Using the formula for

$$\text{the } n\text{th derivative, } f^{(2)}(2) = \frac{3^2}{(2+1)!} = \frac{9}{6} = \frac{3}{2}, \text{ so we know that the second degree term is}$$

$$\frac{3/2}{2!}(x-2)^2 = \frac{3}{4}(x-2)^2 \text{ or } \mathbf{D}.$$

9. Looking at the factorial in the denominator, we see this is most likely related to $\cos x$. Since

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \text{ we can see that there are two things added/changed. The first is that there is a 3 in}$$

the denominator, which means there is a $\frac{1}{3}$ multiplier on the outside (if the 3 was raised to a power, then

it would be inside the $\cos x$ instead of outside). The second is that the x^{2n} has been changed to x^{4n} .

This means that x has been replaced with x^2 . Putting these together gives $\sum_{n=0}^{\infty} \frac{(-1)^n x^{4n}}{3(2n)!} = \frac{1}{3} \cos(x^2)$ or **C**.

10. a) The third degree Taylor polynomial would be setup from the general form with a center at 0, or

$$f(x) \approx P_3(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3. \text{ Plugging in the given values gives us}$$

$$f(x) \approx P_3(x) = 5 + \frac{-3}{1!}x + \frac{1}{2!}x^2 + \frac{4}{3!}x^3 \text{ or } f(x) \approx P_3(x) = 5 - 3x + \frac{1}{2}x^2 + \frac{2}{3}x^3. \text{ Using this, we can find}$$

$$\text{that } f(0.2) \approx P_3(0.2) = 5 - 3(0.2) + \frac{1}{2}(0.2)^2 + \frac{2}{3}(0.2)^3 \text{ or } 4.4253333333.$$

b) The third degree Taylor polynomial would be setup from the general form with a center at 0, or

$$h(x) \approx P_3(x) = h(0) + \frac{h'(0)}{1!}x + \frac{h''(0)}{2!}x^2 + \frac{h'''(0)}{3!}x^3. \text{ So,}$$

$$h(x) = \int_0^{2x} f(t)dt, h'(x) = 2f(2x), h''(x) = 4f'(2x), h'''(x) = 8f''(2x) \text{ [Note: we used the 2}^{\text{nd}} \text{ FTC to}$$

$$\text{find } h'(x)]. \text{ So, } h(0) = \int_0^0 f(t)dt = 0, h'(0) = 2f(0) = 10, h''(0) = 4f'(0) = -12, h'''(0) = 8f''(0) = 8.$$

$$\text{Plugging these in gives us } h(x) \approx P_3(x) = 0 + \frac{10}{1!}x + \frac{-12}{2!}x^2 + \frac{8}{3!}x^3 \text{ or } h(x) \approx P_3(x) = 10x - 6x^2 + \frac{4}{3}x^3.$$

c) From part a), we know that the first 4 terms of $f(x) \approx 5 - 3x + \frac{1}{2}x^2 + \frac{2}{3}x^3$. Since we are adding and subtracting two Maclaurin series that we know, we can write out the first few terms and simplify the like terms to get the first 3 nonzero terms of $k(x)$. So,

$$f(x) \approx 5 - 3x + \frac{1}{2}x^2 + \frac{2}{3}x^3$$

$$\cos x \approx 1 - \frac{x^2}{2} + \frac{x^4}{24}$$

$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$$

but since we are subtracting $\cos x$, it would be

$$f(x) \approx 5 - 3x + \frac{1}{2}x^2 + \frac{2}{3}x^3$$

$$-\cos x \approx -1 + \frac{1}{2}x^2 - \frac{x^4}{24}.$$

$$e^x \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$$

Adding these 3 functions together gives us

$$k(x) = f(x) - \cos x + e^x \approx 5 - 2x + \frac{3}{2}x^2.$$